1. B) Guido van Rossum
2. C) Compulsory Manual Memory Management
3. C) id()
4. B) str
5. C) _my_var
6. B) Data type class of the object
7. B) <class 'float'>
8. D) No keyword is required
9. B) //
10. B) True
11. A) Automatic Reference Counting&Garbage Collection
12. D) Unspecified (unlimited length)
13. B) A namespace is a mapping from names to objects
14. A) True (due to integer caching)
15. B) <class 'list'>
16. D) tuple
17. B) map(int, input().split())
18. B) False
19. B) repr()
20. B) Duck Typing
21. C) elif
22. C) match / case

23. B) _ (underscore)

24. B) B

25. C) Execution simply continues past the match block

26. B) and

27. A) "Even"

28. B) Using pipe: case 1|2:

29. B) Yes, e.g., case x if x > 0:

30. C) "0" (non-empty string)

31. B) Stopping evaluation as soon as outcome is determined

32. C) 10

33. B) "Python"

34. C) Indentation defines code block scope

35. C) Act as a null placeholder statement

36. B) 2, 4, 6

37. B) When the loop completes normally without hitting break

38. C) continue

39. C) Nothing (loop body does not run)

40. B) for i, v in enumerate(items):

41. B) 123

42. C) zip()

43. B) Infinite loop (runs until interrupted)

44. B) 5, 4, 3, 2

45. B) 10

46. D) All of the above

47. A) [0, 2, 4]

48. B) Terminates only the innermost enclosing loop

49. C) range object (iterator/sequence)

50. B) No

51. B) t = (1,)

52. B) Lists are mutable; tuples are immutable

53. B) [1, 2, 3, [4, 5]]

54. C) extend()

55. B) Removes and returns the last element

56. C) [1, 2, 3, 4]

57. B) TypeError: 'tuple' object does not support item assignment

58. B) lst[::-1]

59. A) 1

60. C) O(1)

61. C) Raises ValueError

62. B) lst.sort()

63. C) Only if all elements inside the tuple are immutable

64. B) Assigning multiple values to a tuple and extracting them into variables

65. B) Number of occurrences of element x

66. B) (1, 2, 1, 2)

67. C) Both A and B

68. B) lst.insert(index, val)

69. B) ["c", "a", "t"]

70. A) Shallow copy duplicates references; deep copy duplicates recursively

71. C) Both A and B

72. B) set()

73. B) Must be hashable (immutable)

74. A) d[key] raises KeyError; get() returns None

75. B) An immutable version of a Python set

76. A) Yes, because it is hashable

77. B) {1, 2, 3}

78. B) A|B

79. A) A&B

80. A) {1, 2, 3}

81. B) remove() raises KeyError if x missing; discard() does not

82. B) Guaranteed to be preserved

83. C) d.items()

84. A) {"a": 1, "b": 2}

85. B) Removes and returns the last inserted key-value pair

86. C) Raises AttributeError

87. B) Elements in either set, but NOT in both

88. A) ^ / symmetric_difference()

89. B) 2

90. B) No, because lists are unhashable (mutable)

91. B) d.clear()

92. A) Checks if "a" exists among keys

93. A) {k: v for k, v in iterable}

94. A) A<= B or A.issubset(B)

95. B) |

96. B) Set containing unique characters: {'h', 'e', 'l', 'o'}

97. A) Sets d["c"] = 3 and returns 3

98. C) No, sets are unordered collections

99. C) dict

100. B) Average O(1) time complexity for membership tests via hashing